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AI System Impact Assessment Process

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IAMS/0005	Version 1.0 September 20XX	New documentation	John Doe	Jim Doe	Internal

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Document Control

This document is the responsibility of the Author. It is subject to formal change control after the initial approved release (e.g. version 1.0). Change requests must be communicated to the Author in writing. Agreed changes are recorded in the Version History of the document.

Version History

Date	Version	Author	Changes
11/06/20XX	0.1	John Doe	New Document – initial draft
16/08/20XX	0.2	John Doe	Rewrite following review comments
01/09/20XX	1.0	John Doe	Issued to Business

Reviewers

Name	Role	Date
Jim Doe	AI Ethics Lead	01/09/20XX

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Process

Initiation

- An AI System Impact Assessment must be undertaken when a credible business proposal to develop an AI system is proposed.
- The first impact assessment must be conducted before the final decision to develop the system is made and before policies and guidelines are implemented.
- The purpose of the assessment is to inform the decision and policy. Much of the benefit of an assessment is lost if the assessment is conducted as a fait accompli after the event.
- Impact assessments must be reviewed at different stages of development and use of the AI system and updated as appropriate following changes.
- Impact assessments must be reviewed following significant changes to the subject system. As such, the Change Control Process includes a gate to assess whether a change needs to trigger a review of the associated AI System Impact Assessment.

Investigation

- An AI System Impact Assessment is completed using the AI Impact Assessment Template, derived from the ISO/IES 42005 Standard, through which documents and information that will form the basis of the assessment are recorded.
- The assessment involves discussions with:
  - Data Management Team
  - AI Development Team
  - Quality Assurance Team
  - Cybersecurity Team
  - Risk and Compliance Team
  - AI Research Team
  - Legal Team.
- An AI System Impact Assessment must be led from within the AI Ethics Team. Completed assessments must be reviewed by, as a minimum, the heads of the Data Management, AI Development, Cybersecurity, Legal, Risk and Compliance, AI Research and AI Ethics teams.
- Assessments must receive final approval from the Chief AI Officer.

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Assessment Steps

- The AI System Impact Assessment is divided into the following sections:
  - **Section A** – System Information
    - Provides details of the system being assessed include a high-level description, features, purpose, dependencies on other systems, and intended and unintended uses.
  - **Section B** – Data Information and Quality
    - Provides information about the datasets employed by the system and documentation of the data quality characteristics currently met by the system and any planned to be met in the future.
  - **Section C** – Algorithms and Models Information
    - Provides details of the AI algorithms used in the system and information about the models employed.
  - **Section D** – Deployment Environment
    - Considers the geographical areas in which the system will be deployed, natural languages that the system will need to work with, and complexities and constraints on the deployment of the system.
  - **Section E** – Relevant Interested Parties
    - Records the external and internal interested parties (stakeholders).
  - **Section F** – Actual and Potential Benefits and Harms
    - Identifies the benefits and harms related to the interested parties identified in Section E from the following perspectives:
      - Accountability
      - Transparency
      - Fairness and discrimination
      - Privacy
      - Reliability
      - Safety
      - Explainability
      - Environmental impact
    - It is necessary only to record the harms or benefits to stakeholders for which harms or benefits can be identified; i.e., it is not required to record harms or benefits for every stakeholder/perspective combination.

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- **Section G** – AI system failures and misuse or abuse
  - Identifies the expected impact of anticipated system failure modes and misuse or abuse of the system.

Assessment Approach

- An AI System Impact Assessment follows a similar approach to an audit. It should commence with an opening meeting involving all impacted parties, in which key individuals must be identified from whom information is being gathered.
- Information is gathered in interviews with key individuals, with supporting documentation obtained for review.
- Information gathered is entered into the AI Impact Assessment Template. Supporting documentation can be referenced, or information drawn from supporting documentation can be used to populate the template.
  - *Note that any referenced documentation should include version numbers and issue dates to identify the correct source documents.*
- Upon completion of the template, it must be forwarded to the peer group for review as set out above.
- Upon completion of the peer review, the final draft must be approved for final issue by the Chief AI Officer.
  - *Note that an approval email can be obtained from the Chief AI Officer in place of a physical signature, but a copy of the approval email must be stored with the final version of the AI Impact Assessment.*
- Every effort must be made to ensure the technical accuracy of the AI System Impact Assessment through collaboration with the AI Development and AI Research Teams.
- The AI System Impact Assessment forms part of the system documentation, so needs to be reviewed for consistency with system specification and design documentation.
- It should be noted that AI System Impact Assessments may be shared with external interested parties, such as regulatory bodies and external auditors. Given the sensitive nature of AI System Impact Assessments, they must be marked 'Confidential' and handled in accordance with this classification level.

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